Vrishika V. Mohite

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SUMMARY

Embedded Systems Engineer with 2+ years of experience in firmware validation, system – level debugging, and hardware-software integration for embedded systems. Hands – on experience with C/C++, embedded Linux and RTOS, environments, communication interfaces including CAN, SPI, I2C, and UART, and debugging physical systems using CANalyzer, oscilloscopes, logic analyzers, and JTAG/SWD. Experienced in root-cause analysis, requirements – based validation, and cross – functional debugging in safety – focused engineering environments.

EXPERIENCE

Knorr Bremse India Pvt Ltd **Pune, India**

Embedded Software Engineer (Contract via Shobhi Solutions Pvt Ltd) *Aug 2022 – Jul 2024*

- Diagnosed CAN communication issues using Vector CANalyzer by analyzing message timing, signal values, and bus behavior to isolate firmware, configuration, and interface-related faults during embedded system validation.
- Validated and debugged embedded firmware and hardware-software interactions using oscilloscopes, logic analyzers, and JTAG/SWD tools to investigate timing, interface, and system-level issues on microcontroller-based platforms.
- Authored Software Requirement Analysis methodology, training material, and technical documentation for PTC-Based plugin and reverse-engineering workflows, covering ASPICE SWE.1 requirements analysis, traceability, tool usage , and standardize engineering processes for cross-team knowledge transfer.

San Francisco State University **San Francisco, California**

Teaching Assistant (ENGR 357) *Aug 2025- Jan 2026*

- Assisted in digital logic labs, supporting debugging of combinational and sequential circuits using LTspice.

PROJECTS

Intelligent Posture Monitoring System (STM32MP1 – Cortex-A7/M4) - Developed a heterogeneous embedded system using STM32MP1 with Cortex-M4 real-time firmware and Cortex-A7 embedded Linux. Integrated ToF and IMU sensors over I2C, implemented OpenAMP/RPMsg communication between Linux and real-time firmware, and validated a low-latency 50–100 ms sensor-processing pipeline for real-time posture classification and feedback.

ZonePulse AI – Wearable Heart Rate Monitoring System (ESP32-S3) - Developed a real-time ESP32-S3 wearable monitoring system using layered firmware architecture with HAL and application-level separation. Integrated I2C sensors, SPI display, and PWM feedback, optimizing system response to <250 ms while improving firmware modularity, maintainability, and real-time sensor feedback behavior.

Linux Device Driver for I2C Sensor (Embedded Linux) - Developed a Linux kernel character driver for an I2C-based sensor, including Device Tree configuration, interrupt-driven data handling, and low-level driver interfaces. Performed kernel-level debugging using GDB/LLDB and validated reliable sensor communication for hardware-software integration and real-time data acquisition.

CAN-Based Distributed Embedded Control Network (STM32F4 – FreeRTOS) – Developed a two – node distributed embedded control system using STM32F4 microcontrollers and FreeRTOS, implementing periodic CAN communication, interrupt – driven message handling, node synchronization and timeout-based fault detection. Validated message timing, communication reliability, and faut-recovery behavior using SavvyCAN under normal and simulated error conditions.

Real – Time Scheduling and Fault-Tolerance Frame (C/FreeRTOS) – Developed a real-time embedded software framework in C using FreeRTOS with priority-based task scheduling, inter-task communication, mutexes, watchdog supervision, and fault-recovery mechanisms. Evaluated task latency, deadline behavior, CPU utilization, and priority inversion under varying workloads to analyze real-time system performance and reliability.

SKILLS

Languages: C, Embedded C, C++, Python, Assembly, Bash.

Embedded & Linux Systems: Embedded Linux, RTOS, Bare-Metal Systems, Device Drivers, Device Tree, Cross-Compilation, Root File Systems, Firmware Validation, System Bring-Up, Hardware/Software Integration, Real-Time Sensor Processing

Protocols & Interfaces: I2C, SPI, UART, CAN, USB, GPIO, PWM, DMA, ADC.

Systems & Concepts: Hardware/Software Co-Design, Datasheet-Driven Development, Schematic Review, Register-Level Programming, System-Level Debugging, Code Optimization, Resource-Constrained Embedded Systems Requirements Traceability, Technical Documentation

Tools & Platforms: STM32CubeIDE, Tektronix/Keysight Oscilloscopes, Saleae Logic Analyzer, Vector CANalyzer, JTAG/SWD, ST-Link, MPLAB X, ESP-IDF, Eclipse, JIRA, PTC Integrity, Git, Linux Shell.

Familiarity: Zephyr RTOS, Bluetooth LE fundamentals, bootloader and firmware update concepts, memory maps, CI pipeline concepts.

EDUCATION

San Francisco State University **San Francisco, California**

M.S. in Electrical and Computer *Dec 2026*

- **Concentrations:** Embedded Systems, Intelligent Systems, Automotive Systems
- **GPA:** 3.72 / 4.00
- **Related Coursework:** Embedded Systems Design, ARM Cortex-M/A Architecture, Real-Time Systems & RTOS, Embedded Linux & Device Drivers, Debugging & System Profiling, Machine Learning (Fundamentals), Neural Networks, On-Device (Edge) AI, Neural Machine Interfaces, Signal Processing.

University of Mumbai – Vidyalankar Institute of Technology **Mumbai, India**

B.S. in Electronics and Telecommunication *July 2021*